

2026-05-21 19:30:59.758178: E external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:467] Unable to register cuFFT factory: Attempting to register factory for plugin cuFFT when one has already been registered
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
E0000 00:00:1779391860.174128 57 cuda_dnn.cc:8579] Unable to register cuDNN factory: Attempting to register factory for plugin cuDNN when one has already been registered
E0000 00:00:1779391860.309393 57 cuda_blas.cc:1407] Unable to register cuBLAS factory: Attempting to register factory for plugin cuBLAS when one has already been registered
W0000 00:00:1779391861.307030 57 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
W0000 00:00:1779391861.307108 57 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
W0000 00:00:1779391861.307110 57 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
W0000 00:00:1779391861.307113 57 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
Привіт, Kaggle!
Масив чисел: [1, 2, 3, 4, 5]
Сума чисел: 15

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ПРИКЛАД: XOR для 3-х змінних (рисунок з методички)

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/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:93: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

super().__init__(activity_regularizer=activity_regularizer, **kwargs)

2026-05-21 19:31:45.164222: E external/local_xla/xla/stream_executor/cuda/cuda_platform.cc:51] failed call to cuInit: INTERNAL: CUDA error: Failed call to cuInit: UNKNOWN ERROR (303)

loss 0.3556562066078186

accuracy 0.5

[0 0 0] 0

[0 0 1] 1

[0 1 0] 0

[1 0 0] 0

[0 1 1] 1

[1 0 1] 1

[1 1 0] 0

[1 1 1] 1

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ЗАВДАННЯ: XOR для 4-х змінних

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Таблиця істинності XOR(a, b, c, d):

a b c d | XOR

0 0 0 0 | 0
0 0 0 1 | 1
0 0 1 0 | 1
0 0 1 1 | 0
0 1 0 0 | 1
0 1 0 1 | 0
0 1 1 0 | 0
0 1 1 1 | 1
1 0 0 0 | 1
1 0 0 1 | 0
1 0 1 0 | 0
1 0 1 1 | 1
1 1 0 0 | 0
1 1 0 1 | 1
1 1 1 0 | 1
1 1 1 1 | 0

Навчання моделі (макс. 500 епох)...

loss 0.0872

accuracy 0.9375

Прогнози моделі:

a b c d | Очік. | Прогн.

0 0 0 0 | 0 | 0 ✓
0 0 0 1 | 1 | 1 ✓
0 0 1 0 | 1 | 1 ✓
0 0 1 1 | 0 | 0 ✓
0 1 0 0 | 1 | 1 ✓
0 1 0 1 | 0 | 0 ✓
0 1 1 0 | 0 | 0 ✓
0 1 1 1 | 1 | 1 ✓
1 0 0 0 | 1 | 1 ✓
1 0 0 1 | 0 | 0 ✓
1 0 1 0 | 0 | 0 ✓
1 0 1 1 | 1 | 0 ✗
1 1 0 0 | 0 | 0 ✓
1 1 0 1 | 1 | 1 ✓
1 1 1 0 | 1 | 1 ✓
1 1 1 1 | 0 | 0 ✓

Правильно класифіковано: 15/16

Графік збережено: training_history_xor4.png